

2016.0 RANGE ROVER (LG), 308-07

TRANSFER CASE – VEHICLES WITH: TWIN SPEED TRANSFER CASE

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Transfer Case, refer to the relevant Description and Operation section in the workshop manual. REFER to: [Transfer Case](#) (308-07C Transfer Case - Vehicles With: Twin Speed Transfer Case, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Transfer case ▪ Driveshafts ▪ Differentials ▪ Halfshafts ▪ Fluid leaks ▪ Wheels and tires ▪ Sizes ▪ Pressures ▪ Condition ▪ Wear (tread depth, even/uneven wear)	▪ Fuses ▪ Wiring harnesses and connectors ▪ Transfer case control module ▪ High/low range switch

3. If an obvious cause for an observed or reported concern is found,

correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Warning message/gearbox fault displayed on instrument cluster	Transfer case system fault	- Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Transfer case - Pass - Recheck operation - Fail - Refer to the relevant DTC index
No drive to the front and rear wheels	- Front/rear driveshaft failure/disconnected - Front/center/rear differential failure - Front/rear left/right halfshaft failure/disconnected - Transfer case system failure - Transfer case in Neutral - Transfer case calibration failure	- Check the integrity of the front and rear driveshafts - Check the integrity of the front, center and rear differentials - Check the integrity of the halfshafts - Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index - GO to Pinpoint Test B. - Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement
No drive to the front axle	Transfer case clutch	Check the integrity of the

	worn/burnt	driveshafts, differentials and halfshafts. Rectify as necessary. Check the vehicle history for driveline failures. If a driveline failure has occurred, drain and refill the transfer case fluid, and perform routine - Transfer Case Replacement. Clear the DTCs and re-test. If the fault persists, GO to Pinpoint Test C .
No drive to the front and rear wheels - Automatic transmission gear selection operating correctly	- Transfer case calibration failure - Transfer case solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transfer case actuator position sensor signal A circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transfer case actuator position sensor signal B circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transfer case actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transfer case range change mechanism obstructed - Transfer case range change mechanism not calibrated correctly - Transfer case control	- Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index. Check if the range change actuator is audible during operation. GO to Pinpoint Test B. . Clear the DTCs and retest - GO to Pinpoint Test E. - GO to Pinpoint Test G. - GO to Pinpoint Test H. - GO to Pinpoint Test D. - Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index. Check if the range change actuator is audible during operation. GO to Pinpoint Test B. - Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. Check the transfer case control module for related DTCs and refer to the relevant DTC index. GO to Pinpoint Test B. - Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary

	- module power or ground circuit open circuit, high resistance - Battery/charging system fault	Refer to the relevant section of the workshop manual and check the battery and charging system
Range change inoperative	- Transfer case system fault - Transfer case requires calibration (after installing a new component) - Transfer case too cold - Transfer case control module internal temperature above threshold - Transfer case range change mechanism obstructed - Automatic transmission drag torque excessively high - Transfer case control module power or ground circuit open circuit, high resistance - Battery/charging system fault	NOTES: - Transfer case range changes are inhibited when the transfer case actuator temperature is above 100°C or below -30°C. Only one transfer case range change per minute is permitted when the transfer case actuator temperature is between -30°C and 0°C. - Transfer case range changes are inhibited when the transfer case control module internal temperature is above 100°C. - Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Transfer case - Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement - Drive the vehicle to warm the transfer case and retest. Using the manufacturer approved diagnostic system, check datalogger signal - Motor

		Temperature (0xD11C) - Using the manufacturer approved diagnostic system, check datalogger signals - ECU Internal Temperature (0xD117) - and - Motor Temperature (0xD11C) - and - Motor Over Temperature Events (0xD11F) - Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. If the routine fails, check if the range change actuator is audible during operation. GO to Pinpoint Test B. - GO to Pinpoint Test I. - Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system
Range change, high range to low range, inhibited when the vehicle is moving	Shifting to low range when the vehicle is moving is not possible from 2013 model year	No action necessary
Off road traction reduced and warning message: TRANSMISSION OVERHEAT SLOW DOWN	- Center differential temperature is approaching the overheat threshold - Transfer case fluid level low	NOTE: This warning message may be induced by prolonged off road driving. Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and

		refer to the relevant DTC index. Check datalogger signals - ECU Internal Temperature (0xD117) - and - Motor Temperature (0xD11C) - and - Motor Over Temperature Events (0xD11F) ▪ Check the transfer case for fluid leaks. Refer to the relevant section of the workshop manual and check the transfer case fluid level
Off road traction reduced and warning message: TRANSMISSION FAULT – TRACTION REDUCED	▪ Center differential clutch has failed - Operating as an open differential ▪ Transfer case control module has stopped transmitting on the CAN bus and defaults to open center differential	▪ Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
Judder during low speed manoeuvring with high steering inputs	NOTE: At full lock, there will be some tyre scrubbing; as the temperature drops, scrubbing may be experienced as stick/slip judder. This is normal and does not require any service intervention	▪ Check the tire sizes, pressures and condition ▪ Refer to the relevant section of the workshop manual and drain the transfer case fluid. Flush the transfer case with new fluid and re-fill ▪ Using the manufacturer approved diagnostic system, re-configure the transfer case control module with the latest level software ▪ GO to Pinpoint Test C. ▪ GO to Pinpoint Test A. ▪ Check the integrity of the halfshafts ▪ Refer to the relevant section of the workshop manual and

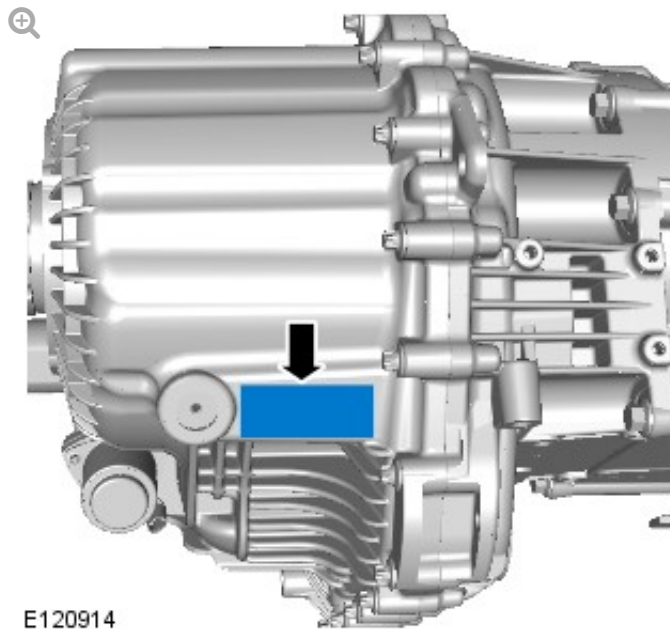
	▪ Excessive difference in tire circumference due to incorrect size, incorrect pressure or wear ▪ Transfer case fluid degraded ▪ Transfer case control module is not configured correctly ▪ Transfer case requires calibration (after installing a new component) ▪ Transfer case clutch not disengaged fully due to water contamination of the transfer case fluid ▪ Front/rear left/right halfshaft failure/disconnected ▪ Rear differential clutch not disengaged fully due to water contamination of the rear differential fluid (active differential)	check the rear differential fluid level/condition
Excessively loud clunk during transfer case range change	▪ Transfer case requires calibration (after installing a new component) ▪ Transfer case control module is not configured correctly	NOTE: During a transfer case range change, a mechanical noise may be audible. This is normal and does not indicate a fault. ▪ Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. Check the transfer case control module for related DTCs and refer to the relevant DTC index. If the fault persists, GO to Pinpoint Test C.

		Using the manufacturer approved diagnostic system, re-configure the transfer case control module with the latest level software
Excessively loud clunk when ignition is set to off (transfer case cycles to rest position)	- Transfer case requires calibration (after installing a new component) - Transfer case control module is not configured correctly - Transfer case clutch worn/burnt	NOTE: During a transfer case range change, a mechanical noise may be audible. This is normal and does not indicate a fault. - Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. Check the transfer case control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, re-configure the transfer case control module with the latest level software - GO to Pinpoint Test C.
Whining noise	Transfer case bearing/chain wear	- Using the manufacturer approved diagnostic system, perform routine - Noise, vibration and harshness sensor diagnostic test - Transfer case - Refer to the relevant section of the workshop manual and install a new transfer case chain - Refer to the relevant section of the workshop manual and install a new transfer case rear output shaft bearing
Transfer case fluid burnt	Excessive off-road	

(darker color than normal)	driving - Transfer case clutch worn/burnt - Electric rear differential fault	NOTE: Some small particles may be present in the transfer case fluid. This is normal and does not indicate a fault. - Refer to the relevant section of the workshop manual and drain and refill the transfer case fluid. Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. Check the transfer case control module for related DTCs and refer to the relevant DTC index - Check the integrity of the driveshafts, differentials and halfshafts. Rectify as necessary. Check the vehicle history for driveline failures. If a driveline failure has occurred, drain and refill the transfer case fluid, and perform routine - Transfer Case Replacement. Clear the DTCs and re-test. If the fault persists, GO to Pinpoint Test C. - Using the manufacturer approved diagnostic system, check the rear differential control module for related DTCs and refer to the relevant DTC index
Noise - Rubbing, scraping or drone. Vibration from front left corner of vehicle. May occur in any gear during steady state driving up to 60 mph (100 km/h), but more prominent in 2nd gear during coast down at approximately 12 mph (20 km/h)	Transfer case wear	If the VIN is earlier than 127697 and the transfer case serial number is lower than A1025861, install a new transfer case

Fluid leak	▪ Transfer case breather pipe blocked ▪ Input shaft seal failure ▪ Front output shaft seal failure ▪ Rear output shaft seal failure ▪ Actuator seal failure ▪ Solenoid seal failure ▪ Mating face seal failure	▪ Check the transfer case breather pipe for blockages ▪ Install a new input shaft seal ▪ Install a new front output shaft seal ▪ Install a new rear output shaft seal ▪ Install a new actuator seal ▪ Install a new solenoid seal ▪ Re-seal the transfer case mating faces
Transfer Case Replacement routine fails repeatedly	▪ Transfer case clutch worn/burnt ▪ Automatic transmission drag torque excessively high ▪ Transfer case control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Check the integrity of the driveshafts, differentials and halfshafts. Rectify as necessary. Check the vehicle history for driveline failures. If a driveline failure has occurred, GO to Pinpoint Test C. ▪ GO to Pinpoint Test I. ▪ Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the relevant section of the workshop manual and check the battery and charging system
Actuator temperature sensor DTC set	▪ Actuator temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ GO to Pinpoint Test F.
Vibration from the transfer case	▪ Installation of incorrect front/rear drive units	▪ Refer to Sections 205-02 and 205-03 for further guidance on how to diagnose and rectify

TRANSFER CASE SERIAL NUMBER LOCATION



There are two numbers stamped onto the transfer case. The first is the part number and the second is the serial number.

TRANSFER CASE ROUTINES

When new transfer case components are installed, the relevant routine must be performed using the manufacturer approved diagnostic system:

- Transfer case control module - Transfer Case Replacement routine
 - Transfer case actuator - Transfer Case Replacement routine
 - Transfer case - Transfer Case Replacement routine
 - Transfer case mode selector solenoid - Solenoid Replacement routine
 - Transfer case mode selector position sensor - Absolute Position Sensor Replacement routine
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DRIVE CYCLE

To validate a transfer case repair, the following drive cycle must be completed:

1. Drive the vehicle a short distance and halt and drive away at least twice
2. Select low range and then select high range (vehicle not moving)
3. Halt the vehicle and select Neutral (manual transmission) or Park (automatic transmission)
4. Set the ignition to off
5. Wait until the transmission control switch Park LED has extinguished (automatic transmission)
6. Start the engine
7. Drive the vehicle until the engine, transmission and transfer case are at normal operating temperature
8. Repeat steps 1 to 5

TRANSFER CASE REPLACEMENT ROUTINE

WARNING:

Stand clear of the vehicle when performing the Transfer Case Replacement routine, as the vehicle may move forward/rearward suddenly. Failure to follow this instruction may result in injury.

NOTE:

When performing the Transfer Case Replacement routine, the transmission should be set to Neutral and the parking brake set to off. Chock the wheels to prevent the vehicle moving excessively (do not apply the brakes as the vehicle must be allowed to move forward/rearward slightly). Failure to follow this instruction will cause the routine to fail.

The Transfer Case Replacement routine should be performed after installing a new transfer case or to calibrate the transfer case mechanism.

TRANSFER CASE CLUTCH CALIBRATION ANGLE RESULTS TABLE

NOTE:

Transfer case clutch calibration angles are displayed as negative values. When assessing the results, the size of the angle should be considered (not the absolute value). For example, -20° is greater than -10° (because the angle is greater).

RESULTS	MEANING	ACTION
Result 1 and Result 2 angles between -20° and -60° and within 10° of each other	Minimal clutch wear (as new)	No action necessary
Result 1 angle or Result 2 angle greater than -170°	Maximum clutch wear	Install a new transfer case
Result 1 angle less than Result 2 angle by 10° or more	Clutch discs burnt	Install a new transfer case and check for driveline failures
Result 1 angle greater than Result 2 angle by 10° or more	Clutch discs damaged due to water ingress	Install a new transfer case

PINPOINT TESTS

NOTE:

Some small particles may be present in the transfer case fluid. This is normal and does not indicate a fault.

PINPOINT TEST A : FLUID CONDITION TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: FLUID CONDITION - PARTICLES	
	1 Drain a sample of fluid from the transfer case.
	2 Check the fluid for solid particles.
	Are any large solid particles present in the transfer case fluid? Yes Install a new transfer case. No GO to Pinpoint Test C .

NOTE:

Perform this test with the engine running and the vehicle stationary.

PINPOINT TEST B : ACTUATOR POSITION TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: ACTUATOR POSITION DATALOGGER SIGNAL	
	1 Using the manufacturer approved diagnostic system, check datalogger signal - Actuator Position (0xD123).
	2 Operate the range change switch to engage high range and low range three times each, and check the Actuator Position value for each event: - High range: ≥ 9 mm - Neutral: = 0 mm

	Low range: ≤ -9 mm
	Does the Actuator Position value match the expected value? Yes No fault found. No Perform routine - Transfer Case Replacement. Clear the DTCs and re-test. If the fault persists, install a new transfer case.

NOTES:

- The clutch calibration angle value can be used to assess the wear state of the transfer case clutch discs.
- The difference between successive clutch calibration angle values can be used to assess the condition of the transfer case clutch discs.
- Transfer case clutch calibration angles are displayed as negative values. When assessing the results, the size of the angle should be considered (not the absolute value). For example, -20° is greater than -10° (because the angle is greater).

PINPOINT TEST C : CLUTCH CALIBRATION ANGLE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: CLUTCH CALIBRATION ANGLE - FIRST READ	
	1 Start the engine.
	2 Using the manufacturer approved diagnostic system, check datalogger signal - Internal Variable Clutch Calibration Angle (0xD902). Record the value (this is Result 1).
	Is Result 1 greater than -170°? Yes Install a new transfer case. No GO to C2.
C2: CLUTCH CALIBRATION ANGLE - SECOND READ	

	NOTE: If the routine fails because the clutch cannot be calibrated, contact Dealer Technical Support. 1 Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement.
	2 Check datalogger signal - Internal Variable Clutch Calibration Angle (0xD902). Record the value (this is Result 2).
	Do Result 1 and Result 2 differ by greater than 10°? Yes Install a new transfer case. No No fault found.

NOTE:

The nominal resistance of the transfer case actuator is 1.0 Ω.

PINPOINT TEST D : TRANSFER CASE ACTUATOR CIRCUIT TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: TRANSFER CASE WIRING HARNESS AND ACTUATOR LOW RESISTANCE	
	1 Set the ignition to off.
	2 Disconnect the largest wiring harness connector from the transfer case control module.
	3 Connect an ohm-meter to terminal 15 and terminal 18 of the transfer case control module wiring harness connector.
	4 Measure the resistance.
	Is the measured resistance less than 0.5 Ω? Yes Resistance too low. GO to D2. No Resistance not too low. GO to D3.
D2: TRANSFER CASE ACTUATOR LOW RESISTANCE	
	1 Disconnect the wiring harness connector from the transfer case

	actuator.
	2 Connect an ohm-meter to terminal 1 and terminal 8 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance less than 0.5 Ω? Yes Resistance too low. Install a new transfer case actuator. No Resistance not too low. Check the transfer case actuator wiring harness for short circuit between the wires. Repair the wiring harness as necessary.

D3: TRANSFER CASE WIRING HARNESS AND ACTUATOR HIGH RESISTANCE

	1 Connect an ohm-meter to terminal 15 and terminal 18 of the transfer case control module wiring harness connector.
	2 Measure the resistance.
	Is the measured resistance greater than 15.0 Ω? Yes Resistance too high. GO to D4. No Resistance not too high. GO to D5.

D4: TRANSFER CASE ACTUATOR HIGH RESISTANCE

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 1 and terminal 8 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance greater than 15.0 Ω? Yes Resistance too high. Install a new transfer case actuator. No Resistance not too high. Check the transfer case actuator wiring harness for open circuit, high resistance. Repair the wiring harness as necessary.

D5: TRANSFER CASE WIRING HARNESS AND ACTUATOR SHORT CIRCUIT TO GROUND

	1 Connect an ohm-meter to terminal 15 of the transfer case control module wiring harness connector and ground.
	2 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes

	No short circuit to ground. No fault found. No Short circuit to ground. GO to D6.
D6: TRANSFER CASE ACTUATOR SHORT CIRCUIT TO GROUND	
	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 15 of the transfer case control module wiring harness connector and ground.
	3 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes Short circuit to ground. Install a new transfer case actuator. No No short circuit to ground. Check the transfer case actuator wiring harness for short circuit to ground. Repair the wiring harness as necessary.

PINPOINT TEST E : TRANSFER CASE SOLENOID CIRCUIT TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: TRANSFER CASE WIRING HARNESS AND SOLENOID LOW RESISTANCE	
	1 Set the ignition to off.
	2 Disconnect the largest wiring harness connector from the transfer case control module.
	3 Connect an ohm-meter to terminal 12 and terminal 9 of the transfer case control module wiring harness connector.
	4 Measure the resistance.
	Is the measured resistance less than 0.2 Ω? Yes Resistance too low. GO to E2. No Resistance not too low. GO to E3.
E2: TRANSFER CASE SOLENOID LOW RESISTANCE	
	1 Disconnect the wiring harness connector from the transfer case solenoid.
	2 Connect an ohm-meter to terminal 1 and terminal 2 of the transfer case solenoid.
	3 Measure the resistance.

	Is the measured resistance less than 0.2 Ω? Yes Resistance too low. Install a new transfer case solenoid. No Resistance not too low. Check the transfer case solenoid wiring harness for short circuit between the wires. Repair the wiring harness as necessary.
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E3: TRANSFER CASE WIRING HARNESS AND SOLENOID HIGH RESISTANCE

	1 Connect an ohm-meter to terminal 12 and terminal 9 of the transfer case control module wiring harness connector.
	2 Measure the resistance.
	Is the measured resistance greater than 10 Ω? Yes Resistance too high. GO to E4. No Resistance not too high. GO to E5.

E4: TRANSFER CASE SOLENOID HIGH RESISTANCE

	1 Disconnect the wiring harness connector from the transfer case solenoid.
	2 Connect an ohm-meter to terminal 1 and terminal 2 of the transfer case solenoid.
	3 Measure the resistance.
	Is the measured resistance greater than 10 Ω? Yes Resistance too high. Install a new transfer case solenoid. No Resistance not too high. Check the transfer case solenoid wiring harness for open circuit, high resistance. Repair the wiring harness as necessary.

E5: TRANSFER CASE WIRING HARNESS AND SOLENOID SHORT CIRCUIT TO GROUND

	1 Connect an ohm-meter to terminal 12 of the transfer case control module wiring harness connector and ground.
	2 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes No short circuit to ground. No fault found. No Short circuit to ground. GO to E6.

E6: TRANSFER CASE SOLENOID SHORT CIRCUIT TO GROUND

	1 Disconnect the wiring harness connector from the transfer case
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	solenoid.
	2 Connect an ohm-meter to terminal 12 of the transfer case control module wiring harness connector and ground.
	3 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes Short circuit to ground. Install a new transfer case solenoid. No No short circuit to ground. Check the transfer case solenoid wiring harness for short circuit to ground. Repair the wiring harness as necessary.

PINPOINT TEST F : TRANSFER CASE ACTUATOR TEMPERATURE SENSOR CIRCUIT TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: TRANSFER CASE WIRING HARNESS AND ACTUATOR TEMPERATURE SENSOR LOW RESISTANCE	
	1 Set the ignition to off.
	2 Disconnect the largest wiring harness connector from the transfer case control module.
	3 Connect an ohm-meter to terminal 5 and terminal 2 of the transfer case control module wiring harness connector.
	4 Measure the resistance.
	Is the measured resistance less than 1 kΩ? Yes Resistance too low. GO to F2. No Resistance not too low. GO to F3.
F2: TRANSFER CASE ACTUATOR TEMPERATURE SENSOR LOW RESISTANCE	
	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 4 and terminal 7 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance less than 1 kΩ? Yes Resistance too low. Install a new transfer case actuator. No

	Resistance not too low. Check the transfer case actuator wiring harness for short circuit between the wires. Repair the wiring harness as necessary.
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F3: TRANSFER CASE WIRING HARNESS AND ACTUATOR TEMPERATURE SENSOR HIGH RESISTANCE

	1 Connect an ohm-meter to terminal 5 and terminal 2 of the transfer case control module wiring harness connector.
	2 Measure the resistance.
	Is the measured resistance greater than 30 kΩ? Yes Resistance too high. GO to F4. No Resistance not too high. GO to F5.

F4: TRANSFER CASE ACTUATOR TEMPERATURE SENSOR HIGH RESISTANCE

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 4 and terminal 7 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance greater than 30 kΩ? Yes Resistance too high. Install a new transfer case actuator. No Resistance not too high. Check the transfer case actuator wiring harness for open circuit, high resistance. Repair the wiring harness as necessary.

F5: TRANSFER CASE WIRING HARNESS AND ACTUATOR TEMPERATURE SENSOR SHORT CIRCUIT TO GROUND

	1 Connect an ohm-meter to terminal 5 of the transfer case control module wiring harness connector and ground.
	2 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes No short circuit to ground. No fault found. No Short circuit to ground. GO to F6.

F6: TRANSFER CASE ACTUATOR TEMPERATURE SENSOR SHORT CIRCUIT TO GROUND

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 5 of the transfer case control

	module wiring harness connector and ground.
	3 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes Short circuit to ground. Install a new transfer case actuator. No No short circuit to ground. Check the transfer case actuator wiring harness for short circuit to ground. Repair the wiring harness as necessary.

PINPOINT TEST G : TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL A CIRCUIT TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
G1: TRANSFER CASE WIRING HARNESS AND ACTUATOR POSITION SENSOR SIGNAL A LOW RESISTANCE	
	1 Set the ignition to off.
	2 Disconnect the largest wiring harness connector from the transfer case control module.
	3 Connect an ohm-meter to terminal 1 and terminal 2 of the transfer case control module wiring harness connector.
	4 Measure the resistance.
	Is the measured resistance less than 16 kΩ? Yes Resistance too low. GO to G2. No Resistance not too low. GO to G3.
G2: TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL A LOW RESISTANCE	
	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 6 and terminal 7 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance less than 16 kΩ? Yes Resistance too low. Install a new transfer case actuator. No Resistance not too low. Check the transfer case actuator wiring harness for short circuit between the wires. Repair the wiring harness as

	necessary.
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G3: TRANSFER CASE WIRING HARNESS AND ACTUATOR POSITION SENSOR SIGNAL A HIGH RESISTANCE

	1 Connect an ohm-meter to terminal 1 and terminal 2 of the transfer case control module wiring harness connector.
	2 Measure the resistance.
	Is the measured resistance greater than 20 k Ω ? Yes Resistance too high. GO to G4. No Resistance not too high. GO to G5.

G4: TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL A HIGH RESISTANCE

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 6 and terminal 7 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance greater than 20 k Ω ? Yes Resistance too high. Install a new transfer case actuator. No Resistance not too high. Check the transfer case actuator wiring harness for open circuit, high resistance. Repair the wiring harness as necessary.

G5: TRANSFER CASE WIRING HARNESS AND ACTUATOR POSITION SENSOR SIGNAL A SHORT CIRCUIT TO GROUND

	1 Connect an ohm-meter to terminal 1 of the transfer case control module wiring harness connector and ground.
	2 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes No short circuit to ground. No fault found. No Short circuit to ground. GO to G6.

G6: TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL A SHORT CIRCUIT TO GROUND

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 1 of the transfer case control module wiring harness connector and ground.

	3 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes Short circuit to ground. Install a new transfer case actuator. No No short circuit to ground. Check the transfer case actuator wiring harness for short circuit to ground. Repair the wiring harness as necessary.

PINPOINT TEST H : TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL B CIRCUIT TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: TRANSFER CASE WIRING HARNESS AND ACTUATOR POSITION SENSOR SIGNAL B LOW RESISTANCE	
	1 Set the ignition to off.
	2 Disconnect the largest wiring harness connector from the transfer case control module.
	3 Connect an ohm-meter to terminal 6 and terminal 2 of the transfer case control module wiring harness connector.
	4 Measure the resistance.
	Is the measured resistance less than 16 kΩ? Yes Resistance too low. GO to H2. No Resistance not too low. GO to H3.
H2: TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL B LOW RESISTANCE	
	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 2 and terminal 7 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance less than 16 kΩ? Yes Resistance too low. Install a new transfer case actuator. No Resistance not too low. Check the transfer case actuator wiring harness for short circuit between the wires. Repair the wiring harness as necessary.

H3: TRANSFER CASE WIRING HARNESS AND ACTUATOR POSITION SENSOR SIGNAL B HIGH RESISTANCE

	1 Connect an ohm-meter to terminal 6 and terminal 2 of the transfer case control module wiring harness connector.
	2 Measure the resistance.
	Is the measured resistance greater than 20 kΩ? Yes Resistance too high. GO to H4 . No Resistance not too high. GO to H5 .

H4: TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL B HIGH RESISTANCE

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 2 and terminal 7 of the transfer case actuator.
	3 Measure the resistance.
	Is the measured resistance greater than 20 kΩ? Yes Resistance too high. Install a new transfer case actuator. No Resistance not too high. Check the transfer case actuator wiring harness for open circuit, high resistance. Repair the wiring harness as necessary.

H5: TRANSFER CASE WIRING HARNESS AND ACTUATOR POSITION SENSOR SIGNAL B SHORT CIRCUIT TO GROUND

	1 Connect an ohm-meter to terminal 6 of the transfer case control module wiring harness connector and ground.
	2 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes No short circuit to ground. No fault found. No Short circuit to ground. GO to H6 .

H6: TRANSFER CASE ACTUATOR POSITION SENSOR SIGNAL B SHORT CIRCUIT TO GROUND

	1 Disconnect the wiring harness connector from the transfer case actuator.
	2 Connect an ohm-meter to terminal 6 of the transfer case control module wiring harness connector and ground.

	3 Measure the resistance.
	Is the measured resistance infinitely high (open circuit)? Yes Short circuit to ground. Install a new transfer case actuator. No No short circuit to ground. Check the transfer case actuator wiring harness for short circuit to ground. Repair the wiring harness as necessary.

PINPOINT TEST I : AUTOMATIC TRANSMISSION DRAG TORQUE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: TRANSFER CASE RANGE CHANGE TEST	
	1 Stop the vehicle on a flat level surface.
	2 Allow the engine to idle.
	3 Set the automatic transmission to Neutral.
	NOTE: The vehicle may creep forward when the brakes are released. Allow the vehicle to move if necessary. 4 Release the brakes.
	5 Attempt a transfer case range change.
	Was the transfer case range change successful? Yes Automatic transmission drag torque excessively high and resisting a transfer case range change when the vehicle is stationary. Refer to the relevant section of the workshop manual and check the automatic transmission. No Automatic transmission drag torque normal.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: [Diagnostic Trouble Code Index - DTC: Transfer Case Control Module](#) (100-00 General Information,

Description and Operation).